

R for absolute beginners

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Workshop structure:

- Short introduction
- List of resources
- RStudio demo
- Presentation and scripts are available on GitHub

What is R?

- R is a programming language for statistical programming and data visualization
- Free, open-source software
- Developed by two professors from the University of Auckland, New Zealand, in the early 1990s
- R has a native command line console, but most people use an integrated development environment, like RStudio

How does R work?

- Basic things to know about R:
 - Everything has a name
 - Everything is an object
 - You do things using functions
 - Functions come in packages

Everything has a name:

- Variables have names, functions have names, your data will have a name
- Names are case sensitive (My_Data is not the same as my_data)
- Follow a naming convention in your code
 - A common one is my_data, my_function, etc
- Do not use spaces, tabs, special characters in your names

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- Objects are how we interact with R
- Objects can store values, text, data tables, figures, etc.
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```
4+5
```

```
nine <- 4+5
```

```
my_name <- "Carolina"
```

Everything is an object:

- Some objects are commonly used across packages: data frames, vectors, lists
- Other object classes are package-specific (common in packages focused on discipline specific applications)

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- Functions perform actions
- Functions produce an output based on the input they receive
- Functions have a parenthesis at the end
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```
my_numbers <- c(1, 2, 3, 4, 5, 6)
```

```
mean(my_numbers)
```

```
[1] 3.5
```

Functions come in packages:

- Functions are bundled into packages or libraries
- Because R is open-source, there are many discipline specific packages
- Saves you from re-inventing the wheel

Functions come in packages:

- CRAN (The Comprehensive R Archive Network) is the main repository of packages
- Packages can be installed with the function `install.packages()` and loaded with `library()`

```
install.packages("phytools")
```

```
library(phytools)
```

```
dplyr::arrange()
```

Base R vs the tidyverse

- Base R is how the basic R packages are usually described
- They don't necessarily share common grammar or data structures
- Tidyverse is a group of packages that share an underlying design philosophy, grammar, and data structures
- Widely used for data science

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```
# in base R, we either state each command or nest them
  summary(my_data)

# tidyverse uses the pipe (%>%) to chain commands
  my_data %>% summary()
```

Some housekeeping tips

- Get in the habit of setting up your working directory in the beginning of each session
- Remember to save your scripts together with any modified data files and figures produced
- Always include comments in your scripts, future you will be thankful
- Keep a running help file with useful commands and tools you use repeatedly
- Popular packages are usually well documented, with tutorials, vignettes, and free resources – use them!
- Finally, be patient with yourself!

Resources

- Instructions to download and install R and Rstudio: <https://www.r-project.org/> and <https://posit.co/downloads>
- CRAN: <https://cran.r-project.org/>
- Hands On Programming with R: <https://rstudio-education.github.io/hopr/>
- R for Data Science: <https://r4ds.hadley.nz/>
- Learning Statistics with R: <https://learningstatisticswithr.com/>
- R Cookbook: <https://rc2e.com/>